

CHEN Yiwei

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Education

National University of Singapore
Ph.D. student

08/2024 – Present
Singapore

- Pursuing a **Ph.D. in Computer Science** under the supervision of Prof. XIAO Xiaokui [🌐 Homepage](#).

Tsinghua University

09/2020 – 06/2024

Double Bachelor's Degree in Science and Engineering

Beijing, China

- B.Sc. in **Mathematics and Physics** & B.Eng. in **Civil Engineering and Systems**.
- GPA: **3.85/4.00**.
- Awards:
 - Distinguished Graduate, Tsinghua University (Top 17%, 2024);
 - Overall Excellence Scholarship, Tsinghua University (Top 20%, 2023);
 - Academic Excellence Scholarship, Tsinghua University (Top 17%, 2023);
 - Overall Excellence Scholarship, Tsinghua University (Top 20%, 2022);
 - Academic Excellence Scholarship, Tsinghua University (Top 17%, 2022).

Research Experience

Federated Learning Contribution Estimation [1]
Database Laboratory, Tsinghua University

09/2022 – 04/2024
Beijing, China

- Supervised by Prof. LI Guoliang [🌐 Homepage](#).
- Key Contributions:
 - An in-depth survey. Examined **29** distinct contribution estimation methods and dissected the problem into three key aspects: data utility metrics, contribution estimation schemes, and optimization techniques.
 - A comprehensive evaluation. Conducted an extensive evaluation of **13** state-of-the-art contribution estimation methods, comparing their effectiveness, robustness, and efficiency, which encompasses 4 datasets, 2 data distributions, and 4 adverse behaviors.
 - An extensive set of observations and findings. Identified the advantages and limitations of different data utility metrics, contribution estimation schemes and optimization techniques across various scenarios, as well as the summarized findings.
 - An extensible testing framework. Developed a flexible testing framework capable of accommodating multiple implemented methods, which serves as a potential benchmark for evaluating performance in this field.

Publications

- [1] Yiwei Chen, Kaiyu Li, Guoliang Li, and Yong Wang. 2024. Contributions Estimation in Federated Learning: A Comprehensive Experimental Evaluation. *Proc. VLDB Endow.* 17, 8 (2024), 2077–2090.

Professional Skills

- Language: English (Professional Working Proficiency), Chinese (Native).
- Technical Skills: Python (PyTorch, Sklearn, Seaborn), C++ (C), MATLAB, SQL; $\text{\LaTeX}$, Drawio (a diagram maker); Git (Github), Overleaf.